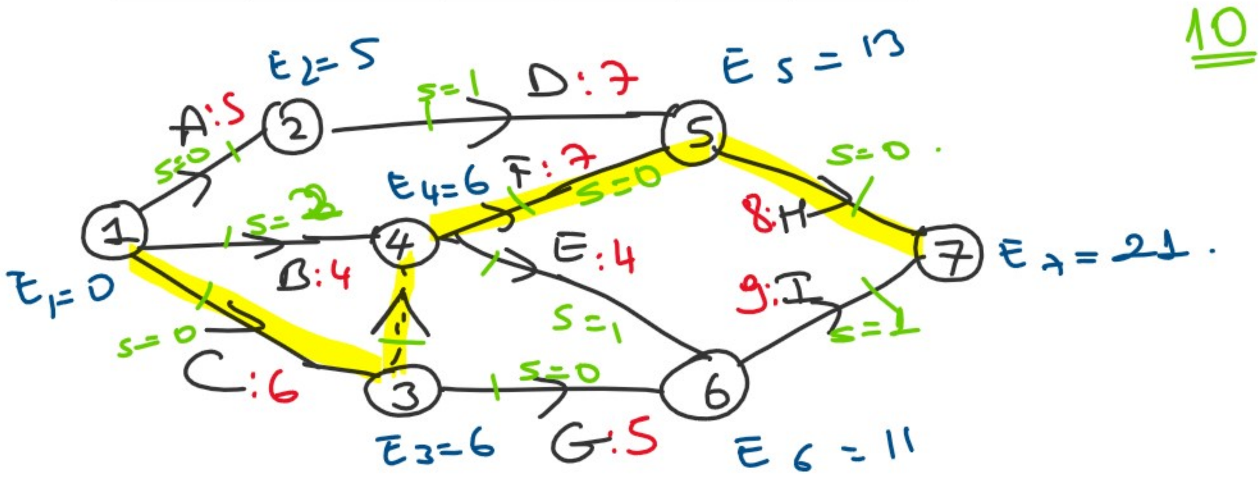
02.09.2021

2 Eylül 2021 Perşembe 15:25

Example2: Table on below gives the activities of a project, their predecessor(s), durations and costs.

a) Draw the network for the project.
b) Determine the critical path.
c) Obtain programs with minimum costs.

Activities	Predecessor(s)	Durations		Costs		
		Normal	Crashed	Normal	Crashed	Slope
A	---	5	3	100	150	25
B	---	4	2	180	200	10
C	---	6	3	170	200	10.
D	A	7	5	200	260	30
E	B, C	4	4	120	120	—
F	B, C	7	7	150	150	—.
G	C	5	3	180	250	35
H	D, F	8	6	200	300	50.
I	E, G	9	6	100	160	20



P<0

C.P : CFH

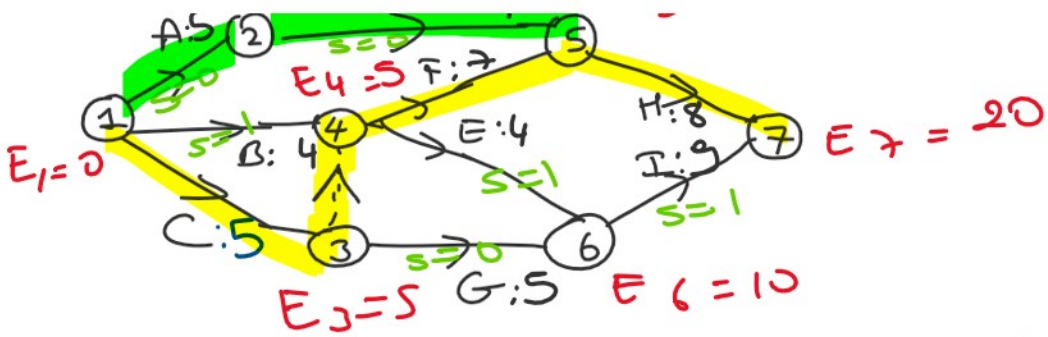
TD : 21

TC : 1400.

C , * , H.

10*, , 50.

C.A of Act C = min { (6-3)/C , 2/B , 0+1/AD } = 1



$\rightarrow$ $\underline{\underline{21-1}}$

PR1

C.P : C~~F~~H :
ADH

T.D : 20

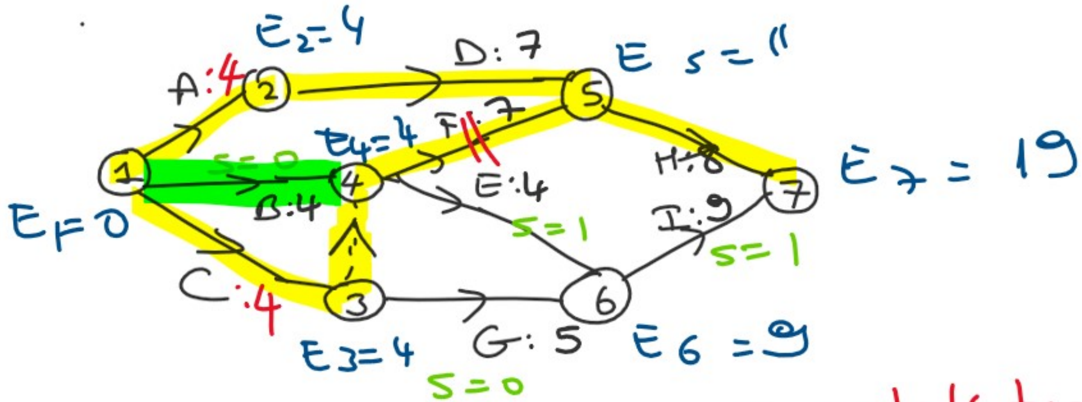
T.C = 1400 + 10 = 1410 .

Possibilities

CA	CD	H
10+25	10+30	50
35	40	

Cr. Am. of A and C $= \min \left\{ \begin{matrix} 5-3, & 5-3, & / & 1 \\ A & C & & B \end{matrix} \right\} = 1$

A , C $\Rightarrow$ 1 unit .



Possibilities

PR2 CP: C~~F~~H
A DH
B ~~F~~H

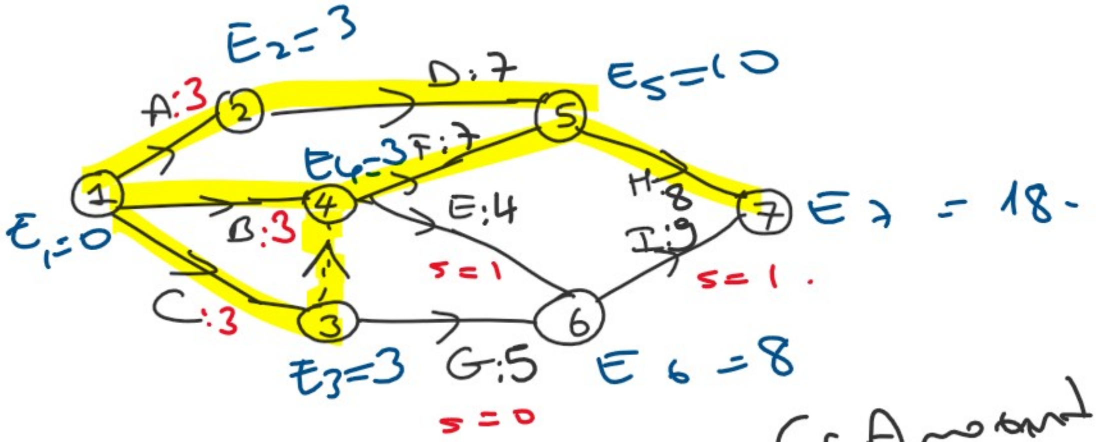
H	CAB	CDB
50	10+25+10	10+30+10
	45	50.

T.D : 19

T.C = 1410 + 35 = 1445 -

Cr. Amount of A, B, C $= \min \left\{ \begin{matrix} 4-3, & 4-2, & 4-3, & / & 1+1, \\ A & B & C & & EI, \end{matrix} \right.$

of
A, B, C = min { A, B, C / EI, GI } = 1 unit.
8+1.

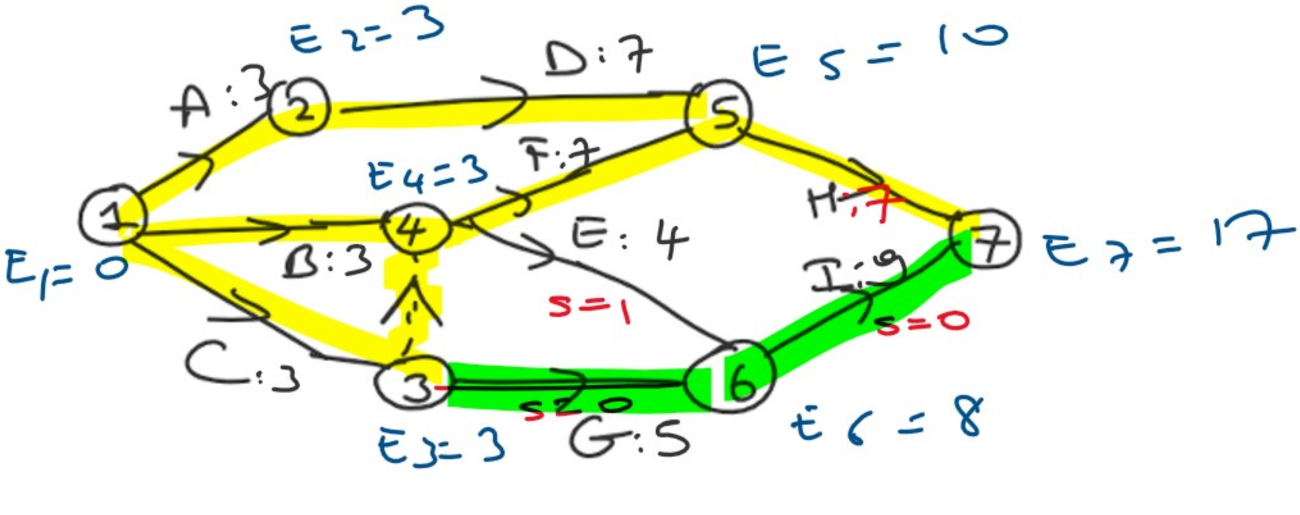


PR3

CP: ~~CF~~H
~~A~~DH
B~~F~~H

$\Longrightarrow$ Cr. Amount of H = min { 8-6, 1, H, EI, GI } = 1

T.D : 18
T.C : 1445 + 45 = 1490.



PR4

C.P. ~~CF~~H
~~A~~DH

Poss.

HG / HI.
50+35 50+20.

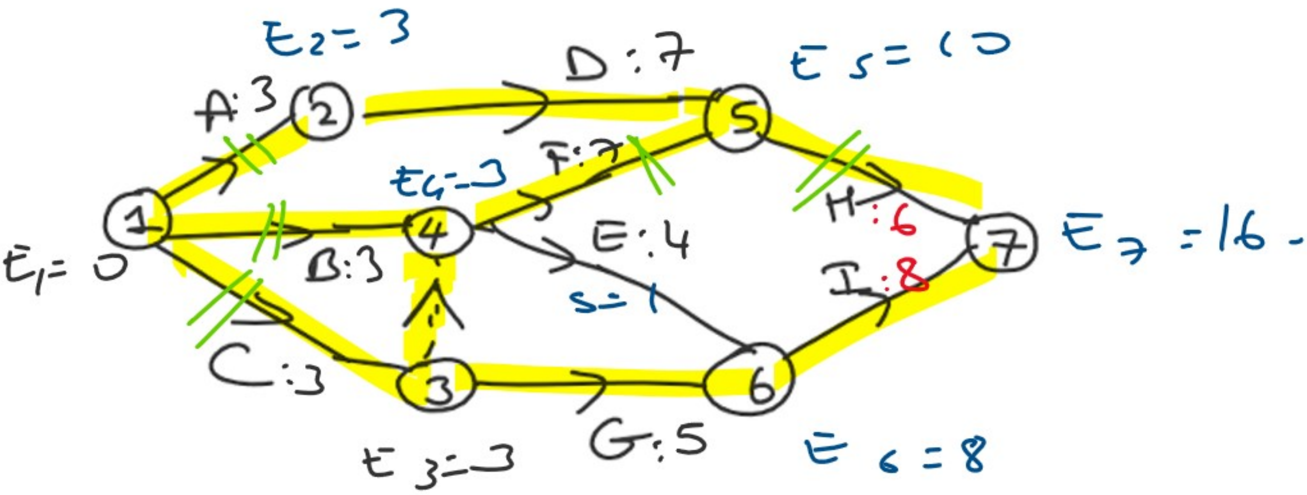
~~A~~ D H
~~B~~ ~~F~~ H
~~C~~ G I

HG / HI.
50+35   50+20.

T.D: 17
T.C: 1490+50 = 1540.

HI → CR. AM. of H and I $= \min \left\{ \begin{matrix} 7-6 & 9-6 \\ H, & I \end{matrix} \Big/ \begin{matrix} 1 \\ E \end{matrix} \right\}$

$= 1.$



P.R.5

C.P. ~~C~~ ~~F~~ ~~H~~ : 16. ⟹ STOP.
~~A~~ D ~~H~~
~~B~~ ~~F~~ ~~H~~
~~C~~ G I

TD: 16
TC: 1540+70 = 1610 //